Entropy-Regularized Adaptive Mixture of Experts for Multi-Output Voice-Based Prediction of Parkinson's Disease Severity

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Abstract

Voice-derived biomarkers provide a non-invasive approach for monitoring Parkinson's disease severity; however, heterogeneous nonlinear relationships between acoustic features and clinical scores limit the effectiveness of single-model prediction strategies. We propose an entropy-regularized adaptive Mixture of Experts (MoE) framework for multi-output prediction of $motor\_UPDRS$ and $total\_UPDRS$ from biomedical voice measurements. Four heterogeneous experts – neural network, random forest, ridge regression, and elastic net – are trained independently and dynamically combined through a lightweight gating network that learns target-specific convex weight distributions per sample. Entropy regularization is incorporated to prevent expert collapse and encourage balanced specialization. The framework is evaluated using frozen train–validation–test splits to ensure strict reproducibility. Experimental results demonstrate that adaptive expert weighting improves generalization compared to individual models and static ensembling approaches, achieving strong predictive performance across both clinical targets while preserving interpretability through analysis of expert contribution patterns. These findings suggest that entropy-regularized adaptive ensembling provides a stable and interpretable strategy for modelling complex biomedical voice signals in Parkinson's disease severity estimation.

Keywords: Parkinson's disease; Voice biomarkers; Disease severity prediction; Mixture of Experts; Multi-output regression; Entropy regularization; Adaptive ensembling

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1. Introduction

Parkinson's disease (PD) is a progressive neurodegenerative disorder characterized by motor and non-motor impairments that significantly affect quality of life. Clinical assessment of disease severity commonly relies on the Unified Parkinson's Disease Rating Scale (UPDRS), which requires in-person evaluation and trained specialists. As PD progresses gradually and heterogeneously across patients, there is growing interest in developing scalable, non-invasive, and objective monitoring strategies that enable continuous assessment outside clinical settings.

Voice analysis has emerged as a promising digital biomarker for PD severity estimation. Motor impairments associated with PD, including hypophonia, dysarthria, and reduced vocal variability, manifest measurably in acoustic features. Prior studies have demonstrated that machine learning models can predict UPDRS scores from biomedical voice measurements with encouraging accuracy. However, the relationship between acoustic features and clinical severity is complex and nonlinear, often varying across individuals and disease stages. Single global models may therefore struggle to capture heterogeneous predictive mechanisms within the data.

Ensemble learning approaches offer a principled way to improve robustness and generalization. In particular, Mixture of Experts (MoE) architectures allow multiple specialized models to contribute dynamically to each prediction. Unlike static averaging ensembles, MoE frameworks learn input-dependent weighting strategies, enabling adaptive specialization across different regions of the feature space. Despite their theoretical advantages, adaptive expert models remain underexplored in biomedical severity estimation tasks.

In this study, we propose an entropy-regularized adaptive Mixture of Experts framework for multi-output prediction of $motor\_UPDRS$ and $total\_UPDRS$ scores from voice-derived features. The proposed system integrates heterogeneous experts – including neural, tree-based, and regularized linear models – and employs a lightweight gating network that learns target-specific convex weight distributions per sample. Entropy regularization is introduced to prevent expert dominance and encourage balanced specialization, enhancing both stability and interpretability.

The contributions of this work are threefold:

(i) formulation of Parkinson's severity estimation as a multi-output adaptive expert learning problem

(ii) integration of entropy-regularized gating to mitigate expert collapse and improve generalization; and

(iii) comprehensive evaluation under frozen data splits to ensure strict reproducibility and transparent comparison against baseline models.

Our results demonstrate that adaptive expert weighting improves predictive consistency while preserving interpretability through analysis of expert contribution patterns, suggesting that entropy-regularized ensembling provides a robust and clinically relevant framework for modeling complex biomedical voice signals.

2. Literature Survey

2.1 Voice Biomarkers for Parkinson's Disease

Voice impairment is among the earliest and most prevalent manifestations of Parkinson's disease (PD), affecting articulation, phonation, and prosody. Acoustic biomarkers derived from sustained phonation and speech recordings - such as jitter, shimmer, harmonic-to-noise ratio, and nonlinear dynamical features—have been widely investigated for PD detection and severity estimation. These studies establish voice analysis as a non-invasive and scalable alternative to clinic-based assessments, motivating the development of computational models for UPDRS prediction.

2.2 Classical Machine Learning Approaches

Early predictive models primarily relied on linear regression, support vector machines, and tree-based methods. Linear and regularized regression techniques offer interpretability but often struggle to model nonlinear relationships inherent in biomedical voice signals. Tree-based approaches such as Random Forest and gradient boosting improve flexibility and robustness, frequently outperforming linear baselines in severity prediction tasks. However, these models remain globally parameterized and may not adapt optimally to heterogeneous patient-specific patterns.

2.3 Deep Learning for Severity Estimation

Neural networks and deep learning models have been introduced to capture complex nonlinear mappings between acoustic features and clinical scores. Multi-layer perceptrons and more advanced architectures demonstrate improved representational capacity, particularly in larger datasets. Nevertheless, deep models can suffer from overfitting, reduced interpretability, and instability in relatively small biomedical datasets. Furthermore, most studies treat severity prediction as either a single-output problem or independently model multiple targets without shared adaptive structure.

2.4 Ensemble Learning and Adaptive Modeling

Ensemble learning techniques have been explored to enhance predictive stability and generalization. Static ensembling methods—such as averaging or stacking—combine multiple models but assign fixed or globally learned weights, limiting their ability to adapt across input regions. Mixture of Experts (MoE) architectures provide a principled alternative by learning input-dependent expert weightings, allowing different models to specialize in distinct subspaces of the data. Despite their theoretical advantages, adaptive expert frameworks remain underexplored in Parkinson's severity modeling, particularly for simultaneous multi-output prediction.

2.5 Identified Research Gap

Existing studies demonstrate that (i) nonlinear models outperform linear approaches, (ii) ensemble strategies improve robustness, and (iii) voice-derived features carry meaningful severity information. However, three limitations persist:

(i) The absence of adaptive, sample-specific expert weighting mechanisms;

(ii) The limited exploration of joint multi-output severity modeling (i.e., $motor\_UPDRS$ and $total\_UPDRS$ predicted simultaneously);

(iii) The lack of regularization strategies to mitigate expert dominance and collapse in adaptive ensemble frameworks.

To address these gaps, we propose an entropy-regularized adaptive Mixture of Experts framework that dynamically combines heterogeneous predictors through target-specific gating while encouraging balanced expert specialization. This approach integrates the strengths of linear, tree-based, and neural models within a unified, interpretable, and reproducible architecture.